SHAO-CHUN HUNG

Computer Vision | Machine Learning | Applied AI

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Summary

Graduate researcher in Mathematical Modeling and Scientific Computing at NYCU, with hands-on experience in machine learning, computer vision, data analysis, and AI workflow prototyping. Built end-to-end project pipelines spanning data preprocessing, modeling, evaluation, and experimentation, from YOLO-based ship detection to graph-based MRT passenger flow analysis.

Skills

Programming | Python · Matlab · Git · SQL

AI / Modeling | ML · PyTorch · Computer Vision · YOLO · OpenCV

LLM / Workflow | Prompt Engineering · AI-Agents-Assisted Workflow

Data Analysis | Data Visualization · Pandas · Numpy

Tools | GitHub · Docker · API Integration

Research & Projects

Optimizing Ship Detection in Multi-Scale Environments

NYCU
03/2025 - Present

- Developed a ship detection pipeline for multi-scale environments and evaluated model performance across complex visual scenes.
- Improved the YOLO11 architecture by comparing feature extraction modules by using Python and PyTorch.
- Built a GUI tool to support image enhancement, annotation, copy-paste augmentation, and post-training prediction analysis for the detection workflow.
- Achieved a +1.3 mAP@50 improvement over baseline and provided structured analysis across vessel categories, inference speed, and detection accuracy.

Passenger Flow Analysis of the Taipei MRT

NTNU
10/2022 - 07/2023

- Built a passenger flow analysis project for the Taipei MRT using open government transportation data.
- Constructed a 119 × 119 station transfer matrix and applied Markov Chain modeling, PageRank, and Power Method iteration.
- Identified high-mobility interchange hubs and surfaced regional transit patterns through algorithmic analysis.

CPBL Batting and Pitching Performance Analytics

NTNU
10/2021 - 01/2022

- Built a structured analytics project for CPBL batting and pitching performance using multi-season official statistics.
- Parsed and cleaned raw web data, then designed a cross-year comparison pipeline with conditional filtering rules in Python.
- Identified recurring performance patterns across three consecutive championship-winning teams and delivered end-to-end analytical reporting.

Awards

1st Place — 2026 TSMC IT CareerHack

- Worked on legacy code refactoring and helped establish a more verifiable and reproducible refactoring process.
- Designed and optimized spec-driven prompts to improve task alignment and output structure.
- Converts C to Rust and migrates MVC to decoupled architecture, reducing refactoring cycle time by ~80%.

Best Research Award — 2022 Applied Mathematics Conference

- Led project execution : mathematical model derivation, Python implementation, and results interpretation.
- Helped address trip-planning challenges for tourists by revealing major interchange hubs and passenger flow patterns

Experience

Teaching Asssistant, **Calculus** | NYCU

09/2024 - Present

- Developed an **AI-agent-assisted workflow** to support student registration and improve class administration efficiency.
- ***Awarded Best Teaching Assistant of Academic Year 2025***

Education

National Yang Ming Chiao Tung University (NYCU)

Hsinchu, Taiwan
09/2024 - Present

M.S. — Mathematical Modeling and Scientific Computing

National Taiwan Normal University (NTNU)

Taipei, Taiwan
09/2020 - 06/2024

B.S. — Mathematics

- GPA：**4.0/4.3** | Department Rank：**Top 5**